

The Erosion of the Majority

Cognitive Outsourcing, Practised Competence, and the Population-Level Preservation Problem

Anthony Paterson

Fractal Media Infrastructure

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Abstract

Artificial intelligence is frequently discussed in terms of productivity, employment displacement, dependency, and the changing distribution of human and machine labour. Less attention has been given to a longer-term possibility: that increasingly capable cognitive automation may alter not merely which tasks humans perform, but which cognitive capabilities remain broadly practised across a population.

This paper distinguishes **cognitive capacity** from **practised competence**. Humans need not become biologically less intelligent for historically ordinary intellectual capabilities to become increasingly uncommon. Skills may remain technically preserved in archives, institutions, specialist communities, or small populations of enthusiasts while becoming functionally absent at population scale. Historical transitions in craft and industrial production provide a partial analogy: capabilities once widely distributed can become specialised, artisanal, reconstructive, or antiquarian when technological systems remove the practical necessity of maintaining them.

Artificial intelligence extends this dynamic into an unusually broad domain. Writing, programming, research synthesis, design, planning, mathematical work, administration, argumentation, technical diagnosis, and even task decomposition itself are increasingly amenable to delegation through a common technological substrate. This creates the possibility of simultaneous cross-domain erosion rather than the displacement of one bounded craft.

The central risk is therefore not that artificial intelligence necessarily reduces human intelligence. It is that widespread cognitive outsourcing may weaken the social, educational, and economic mechanisms through which intelligence is routinely exercised and converted into competence. If direct participation becomes unnecessary for obtaining valued outcomes, the majority may gradually cease developing capabilities that remain biologically available to them.

This paper frames that possibility as a problem of **population-level capability preservation**. It argues that preserving records of knowledge is insufficient if living competence becomes rare, and that human-AI systems should be evaluated partly by whether they preserve meaningful pathways for human participation, understanding, and independent execution.

Keywords: cognitive outsourcing; automation; skill decay; cognitive offloading; human-AI collaboration; competence preservation; technological dependence

1. Introduction

Human history contains many examples of technologies replacing forms of labour that were once ordinary.

Skills that previously belonged to everyday economic life can, over time, become specialised crafts. Hand weaving, leatherworking, blacksmithing, typesetting, shoemaking, furniture making, food preservation, navigation, and numerous manufacturing techniques have undergone versions of this transition. Knowledge often survives. Practitioners may remain. Museums, specialist workshops, historians, reconstruction projects, and enthusiast communities may preserve substantial parts of a tradition.

Yet this survival should not be confused with continuity at the population level.

A society in which ten thousand people practise a skill is materially different from one in which ten million do, even if both societies can truthfully claim that the skill still exists.

The distinction becomes more consequential when the activity being displaced is cognitive rather than narrowly physical.

Contemporary artificial intelligence systems increasingly mediate writing, programming, research, illustration, planning, administration, mathematical work, design, business development, translation, teaching, technical troubleshooting, and decision support. Agentic systems extend this capacity further by decomposing goals, selecting intermediate tasks, evaluating outputs, revising approaches, and coordinating additional systems.

This raises a question that employment-centred analyses do not fully capture:

What happens when technologies do not merely perform human tasks, but increasingly remove the practical necessity for humans to develop the competencies required to perform those tasks themselves?

The argument of this paper is not that humans will become biologically less intelligent.

Nor is it that all automation is harmful, that manual execution is inherently virtuous, or that technologically mediated cognition should be rejected.

The concern is instead about **application**.

Human intelligence becomes socially useful through training, repetition, cultural inheritance, education, apprenticeship, experimentation, failure, correction, and repeated participation in difficult processes. A population can retain the biological capacity for a skill while losing much of the practised competence associated with it.

Artificial intelligence makes this possibility unusually significant because it can potentially mediate many domains at once.

1.1 Positioning in Existing Literature

The argument developed here sits alongside several established research traditions without being reducible to any one of them. Research on **cognitive offloading** examines how people use external resources to reduce internal cognitive demand and how such offloading changes what is remembered or internally processed (Risko & Gilbert, 2016; Sparrow, Liu, & Wegner, 2011). Research on automation has long examined the risks of over-reliance, passive monitoring, and operators becoming “out of the loop” when active involvement is reduced (Bainbridge, 1983; Endsley & Kiris, 1995; Parasuraman & Riley, 1997). Skill-retention research likewise shows that nonuse can produce substantial decay in trained knowledge and performance over time (Arthur et al., 1998).

A second relevant tradition concerns the social and tacit character of competence. Situated-cognition research emphasises that knowledge develops through activity and context rather than existing only as detached information (Brown, Collins, & Duguid, 1989). Work on tacit knowledge similarly distinguishes explicit records from forms of know-how that are embodied, relational, or collectively sustained through practice (Collins, 2010). These traditions support a distinction central to this paper: preserving information about a capability is not necessarily the same as preserving a population that can exercise it.

The contribution here is therefore conceptual and scalar. It does not claim that current AI use has already produced population-wide cognitive decline. Instead, it asks what follows if mechanisms already studied at the level of individual offloading, automation, nonuse, and situated practice begin operating simultaneously across many cognitive domains through a common technological substrate. The resulting question is one of distribution: how much living, independently exercisable competence remains across a population when high-quality outcomes can increasingly be obtained without direct participation in the processes that historically developed that competence?

2. Capacity Is Not Competence

The distinction between cognitive capacity and practised competence is fundamental.

The argument does not require any claim that present-day humans possess greater biological intelligence than earlier humans. What differs enormously across societies and historical periods is the cultural, technological, institutional, and educational environment in which people develop. Situated-cognition research likewise stresses that knowledge and competence are shaped by activity, context, and culture rather than by abstract capacity alone (Brown, Collins, & Duguid, 1989).

A person raised within a Neolithic community would acquire a repertoire suited to that environment: toolmaking, navigation, seasonal knowledge, animal behaviour, resource identification, social obligations, food production, environmental interpretation, and other forms of practical competence unfamiliar to most contemporary people.

A modern person instead inherits literacy, numeracy, formal education, digital systems, specialised institutions, accumulated science, mass infrastructure, and thousands of years of recorded knowledge.

The underlying distinction is not simply one of intelligence. It is one of **what intelligence has been trained to do**.

This matters because many public discussions of cognitive automation collapse two very different propositions:

humans may become less intelligent;

and

humans may practise some forms of intelligence less frequently.

The first is a claim about biological or general cognitive capacity.

The second is a claim about cultural practice.

The second requires no biological decline whatsoever.

A person may remain fully capable of learning long-form writing while never developing the ability to write independently. A programmer may retain the capacity to understand software architecture while rarely exercising it because systems increasingly generate, test, revise, and document implementations automatically. A researcher may retain the cognitive potential required for literature synthesis while delegating retrieval, comparison, argument construction, and drafting to agents.

The brain has not necessarily changed.

The **practised repertoire** has.

This is analogous to physical craft. Modern populations contain enormous numbers of people biologically capable of learning leatherworking. Very few possess the competence to construct a saddle.

The capability remains latent.

The practice has become uncommon. This distinction is consistent with research showing measurable decay after nonuse and with accounts of tacit knowledge that cannot be reduced to stored instructions alone (Arthur et al., 1998; Collins, 2010).

3. From Everyday Competence to Artisan Survival

Industrialisation provides an imperfect but useful historical comparison.

When machines make a form of production substantially cheaper, faster, or more scalable, the displaced skill seldom disappears immediately. Its social position changes first.

An ordinary trade may become a specialised trade. A specialised trade may become a premium craft. A premium craft may eventually survive primarily through enthusiasts, heritage industries, reconstruction specialists, museums, or bespoke markets.

The knowledge is not necessarily destroyed.

It becomes **thinly distributed**.

Leather goods offer a straightforward example. Industrial production does not prevent somebody from learning to cut, stitch, finish, and construct leather products by hand. Indeed, contemporary craftspeople may have access to tools and information unavailable to historical practitioners.

But mass production removes the ordinary economic necessity for most people to possess those capabilities.

The result is not universal incapacity. It is minority preservation.

This distinction produces an important principle:

Knowledge can survive while population competence erodes.

The existence of specialist practitioners does not mean a society retains the capability in any broad sense.

Nor does documentation solve the problem completely.

Historical reconstruction illustrates the limitations of archived knowledge. Instructions may preserve explicit procedure while omitting tacit understanding: how a material should feel, what a process should sound like, when an apparently minor variation matters, what experienced practitioners recognise without conscious articulation, or what nobody bothered documenting because it was considered obvious. The broader literature on tacit knowledge provides a well-established basis for distinguishing codified information from know-how sustained through practice (Collins, 2010).

The statement “the knowledge was written down” can therefore conceal the disappearance of the living practice required to interpret and apply it.

4. Artificial Intelligence Changes the Scale of the Analogy

The industrial analogy becomes more concerning when applied to general-purpose cognitive technologies.

A power loom automates weaving.

It does not simultaneously automate legal drafting, software engineering, mathematical derivation, historical research, illustration, business planning, education, translation, scientific synthesis, and administrative reasoning.

Artificial intelligence increasingly spans all of these domains through one broad technological layer.

This creates the possibility of **cross-domain capability erosion**.

A society may historically lose widespread competence in one craft while retaining enormous amounts of cognitive activity elsewhere. A person who no longer produces textiles may still plan businesses, perform

mathematics, teach apprentices, write correspondence, troubleshoot machinery, participate in institutions, and reason through unfamiliar problems.

General-purpose AI potentially reaches into the processes underlying all of those activities.

This is why the concern cannot be reduced to a collection of separate occupational substitutions.

Writing, mathematics, programming, research, design, and planning differ in content, but their practice repeatedly exercises overlapping cognitive processes: decomposition, abstraction, persistence through uncertainty, error recognition, comparison, revision, causal reasoning, model construction, and judgment.

If these activities are increasingly delegated together, the displaced element may not merely be a collection of isolated skills.

It may include portions of the **practice environment through which higher-order competence is developed**.

5. The Shortcut to the Shortcut

People routinely use tools and external resources to reduce effort, memory load, and other cognitive demands. This behaviour is neither pathological nor surprising; cognitive offloading is a normal feature of human activity (Risko & Gilbert, 2016). Technologies are often adopted because they reduce physical labour, cognitive burden, time, cost, or uncertainty.

Historically, however, many shortcuts still left the user inside a larger process.

A calculator removes arithmetic effort, but somebody must ordinarily decide what calculation matters.

A power tool reduces physical work, but somebody must decide what is being built.

Search engines dramatically reduce retrieval costs, but users still commonly interpret the results.

Contemporary agentic systems increasingly permit another step.

The system may be asked not merely to perform a task, but to determine the tasks required to satisfy a desired outcome.

Consider the difference between:

Write this product description.

and:

Build me a successful business.

The latter request potentially delegates customer selection, problem definition, product design, pricing, branding, marketing, operational planning, metrics, research, implementation, and evaluation.

The human has not merely shortened a process.

The human may have **shortcutted participation in determining the process itself**.

This produces an unusual incentive structure.

For much of history, obtaining sophisticated outputs often required becoming more capable.

If a person wanted to construct a complex argument, build software, perform analysis, or establish a specialised enterprise, some degree of competence was instrumentally necessary.

Increasingly capable AI weakens that coupling.

A person may be able to obtain the outcome without developing much of the competence historically required to produce it.

The shortcut therefore begins to encompass the mechanism by which humans previously learned to create the outcome.

6. The Erosion of the Majority

This paper calls the resulting population-level phenomenon **the erosion of the majority**.

The term does not imply that all competence disappears.

On the contrary, many capabilities may survive extremely well within specialised communities.

There may continue to be programmers who write software directly, mathematicians who derive results independently, writers who construct prose unaided, craftspeople who deliberately preserve traditional processes, and educators who teach foundational methods.

The relevant question is whether these remain **ordinary capabilities distributed across substantial populations**, or become increasingly specialist ones.

Suppose that in a future society nearly everyone retains the biological capacity to perform complex mathematical reasoning, but the overwhelming majority relies on automated systems for mathematical work.

A small number of universities, enthusiasts, historians, and specialists continue practising independent mathematics.

Has mathematical knowledge disappeared?

Clearly not.

Has the population retained mathematical competence in the same sense as a society in which millions routinely exercise it?

Also clearly not.

At some threshold, technically surviving competence becomes practically negligible at population scale.

This is the distinction between **archival survival** and **living distribution**.

A civilisation cannot meaningfully claim broad possession of a capability merely because a sufficiently determined specialist can reconstruct it.

7. Academia Is Not Outside the System

It may be tempting to imagine universities and research institutions as permanent repositories that will preserve displaced cognitive capabilities.

That assumption is weak.

Academia is composed of the same populations exposed to the same technological incentives.

Academic practice also changes generationally.

Senior researchers retire and die. New cohorts replace them. Practices initially regarded with suspicion become tolerated, then normalised, then assumed. Skills once defended as foundational can eventually be questioned on the grounds that contemporary professional practice no longer requires them.

The following is a plausible institutional pathway rather than an empirical forecast. It need not involve explicit abandonment:

students must perform this themselves;

becomes

students should understand it, although automated assistance is acceptable;

which becomes

the important skill is supervising automated systems;

which may eventually become

why devote substantial curriculum time to a procedure almost nobody performs directly?

Every transition can be locally reasonable.

Across generations, however, the accumulated result may be substantial.

This presents a preservation paradox.

A future academic institution might retain enormous archives describing earlier modes of human reasoning while possessing progressively fewer researchers capable of reproducing those modes without the same automated infrastructure whose effects they are attempting to study.

In the extreme case, researchers might employ AI systems to reconstruct how humans once performed cognitive work without AI systems.

The knowledge survives.

The independent epistemic position from which to validate the reconstruction may not.

8. The Loss of the Ladder

A useful metaphor is that civilisation provides each generation with a ladder.

Nobody individually recreates mathematics, science, literature, engineering, law, or philosophy from first principles. Each generation inherits accumulated knowledge and climbs from a starting point produced by earlier generations.

Artificial intelligence can make the ladder dramatically easier to climb.

That is potentially enormously beneficial.

The concern begins when the technology no longer merely assists climbing but increasingly makes climbing unnecessary.

If outputs can be accessed through infrastructure without developing the associated competence, then the ladder may remain physically present while fewer people use it.

Over time, neglected rungs become culturally unfamiliar.

Teaching shifts.

Institutional expectations change.

Economic incentives disappear.

Eventually, independent competence can begin to resemble traditional craft: respected, occasionally useful, deliberately preserved, but no longer ordinary.

At that stage the relevant question is not whether civilisation still owns the ladder.

It is whether enough people still know how to climb it.

9. Why Preservation Cannot Mean Refusing Automation

The argument presented here should not be mistaken for technological nostalgia.

Human societies routinely benefit from removing unnecessary labour.

There is no intrinsic virtue in manually performing every arithmetic operation, copying every document by hand, or refusing tools that increase productivity.

Nor is widespread expertise in every domain desirable or possible.

The relevant distinction is between **automation that amplifies competence** and automation that progressively removes the need to develop competence. Classic automation research already shows that the allocation of function between human and machine can alter monitoring, situation awareness, and the ability to resume direct control (Bainbridge, 1983; Endsley & Kiris, 1995; Parasuraman & Riley, 1997).

Human-AI collaboration can substantially increase what individuals are capable of producing while retaining human participation in framing, architecture, judgment, evaluation, and revision.

A person may delegate implementation while retaining understanding of the system being implemented.

A researcher may use AI for retrieval while retaining authorship of the argument.

A builder may use an agent against a bounded specification whose goals, constraints, and architecture remain human-directed.

Such workflows do not preserve every historical form of labour.

Nor should they.

What they preserve is a meaningful coupling between **human understanding and technological output**.

The alternative is not automatically harmful. There will be legitimate contexts in which complete automation is desirable.

The concern arises when complete delegation becomes the cultural default across domains in which widespread human competence remains socially valuable.

10. Capability Preservation as a Social Problem

If this risk is taken seriously, preservation cannot consist solely of storing knowledge.

A library is not a population.

A source-code archive is not a programming culture.

A mathematics textbook is not a community of mathematically competent people.

Preserving capability requires continued practice.

This suggests a distinction between **knowledge preservation** and **competence preservation**.

Knowledge preservation stores information that describes how something may be done.

Competence preservation maintains sufficient populations who actually do it, teach it, correct errors in it, transmit tacit knowledge, and remain capable of evaluating both human and automated performance. This emphasis on practice is supported by both skill-retention research and accounts of tacit knowledge (Arthur et al., 1998; Collins, 2010).

The scale required will differ across domains. No plausible society needs everyone independently capable of advanced mathematics, compiler construction, historical brewing, or structural engineering.

But neither is a society resilient merely because thirty specialists somewhere retain each capability.

The problem is one of distribution.

How much independent competence must remain distributed across a population for that society to meaningfully retain agency over the systems upon which it depends?

This question is likely to become increasingly important as AI systems mediate larger portions of cognitive production.

11. Relationship to the Vanishing Middle

The erosion of the majority is closely related to a second phenomenon: the disappearance of intermediate human-AI workflows.

If cultural expectations polarise toward either direct manual work or extensive autonomous delegation, collaborative workflows in which humans retain substantive authorship while machines amplify execution may become less visible.

This matters because the middle provides one potential mechanism for retaining competence while capturing substantial benefits from automation.

A worker who frames the problem, defines the architecture, constructs bounded instructions, evaluates intermediate outputs, and revises the process remains cognitively engaged in ways that differ meaningfully from a worker who supplies only a desired outcome.

The distinction is not between “using AI” and “not using AI.”

It is between different **allocations of cognition** within human-machine systems.

If the middle collapses culturally, population-level competence erosion may accelerate because fewer people encounter workflows in which direct understanding remains instrumentally useful.

The two problems therefore reinforce one another.

As competence becomes less common, deep collaborative participation becomes harder.

As deep collaborative participation becomes less common, competence receives less practice.

11.1 Empirical Questions

The framework is intended to generate testable questions rather than to substitute for empirical evidence. Relevant work could examine whether sustained AI delegation predicts measurable changes in unaided performance; whether different human-AI workflow designs produce different retention outcomes; whether competence lost through nonuse can be reacquired quickly; which domains are most vulnerable to thinning at population scale; and what level of distributed competence is sufficient for institutional resilience, independent evaluation, and recovery after system failure.

Longitudinal designs will be especially important. Short-term productivity gains and short-term learning outcomes cannot by themselves establish whether a population is preserving or eroding independent competence across years or generations. The central claim of this paper is therefore not that erosion is already demonstrated, but that population-level distribution should become an explicit object of measurement before increasingly complete delegation makes the underlying change difficult to observe.

12. Conclusion

The long-term cognitive risk of artificial intelligence may not resemble a dramatic decline in human intelligence.

A quieter outcome is possible.

Humans may remain biologically capable of extraordinary reasoning while exercising some forms of that reasoning less frequently.

Skills may survive while becoming thinly distributed.

Knowledge may remain perfectly archived while living competence becomes rare.

Institutions may continue functioning while increasingly depending upon infrastructure that fewer people can independently reproduce, inspect, or challenge.

History provides many examples of once-ordinary skills becoming specialist crafts after technological substitution. Artificial intelligence differs in the breadth of its potential reach. It can mediate not merely one trade or production method but many of the cognitive activities through which competence itself is formed.

The resulting concern is not that automation inevitably produces cognitive decline.

Nothing about this trajectory is inevitable.

But strong incentives point toward greater delegation: reduced effort, increased output, lower cost, and increasingly capable systems. Countervailing practices - education, professional norms, human-AI collaboration, direct practice, and deliberate competence preservation - require active maintenance.

This asymmetry deserves attention.

A society may retain access to enormous machine capability while gradually reducing the proportion of humans who independently understand substantial parts of what that capability produces.

The challenge is therefore not to prevent technological shortcuts.

It is to avoid allowing those shortcuts to remove, at population scale, the pathways through which humans learn to understand and act without them.

The objective is not preservation for nostalgia's sake.

It is to ensure that human capability remains **living rather than archival, distributed rather than ceremonial, and practised rather than merely possible**.

References

Arthur, W., Jr., Bennett, W., Jr., Stanush, P. L., & McNelly, T. L. (1998). Factors that influence skill decay and retention: A quantitative review and analysis. *Human Performance*, 11(1), 57-101.
https://doi.org/10.1207/s15327043hup1101_3

- Bainbridge, L. (1983). Ironies of automation. *Automatica*, 19(6), 775-779. [https://doi.org/10.1016/0005-1098\(83\)90046-8](https://doi.org/10.1016/0005-1098(83)90046-8)
- Brown, J. S., Collins, A., & Duguid, P. (1989). Situated cognition and the culture of learning. *Educational Researcher*, 18(1), 32-42. <https://doi.org/10.3102/0013189X018001032>
- Collins, H. (2010). *Tacit and explicit knowledge*. University of Chicago Press. <https://doi.org/10.7208/chicago/9780226113821.001.0001>
- Endsley, M. R., & Kiris, E. O. (1995). The out-of-the-loop performance problem and level of control in automation. *Human Factors*, 37(2), 381-394. <https://doi.org/10.1518/001872095779064555>
- Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, 39(2), 230-253. <https://doi.org/10.1518/001872097778543886>
- Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences*, 20(9), 676-688. <https://doi.org/10.1016/j.tics.2016.07.002>
- Sparrow, B., Liu, J., & Wegner, D. M. (2011). Google effects on memory: Cognitive consequences of having information at our fingertips. *Science*, 333(6043), 776-778. <https://doi.org/10.1126/science.1207745>